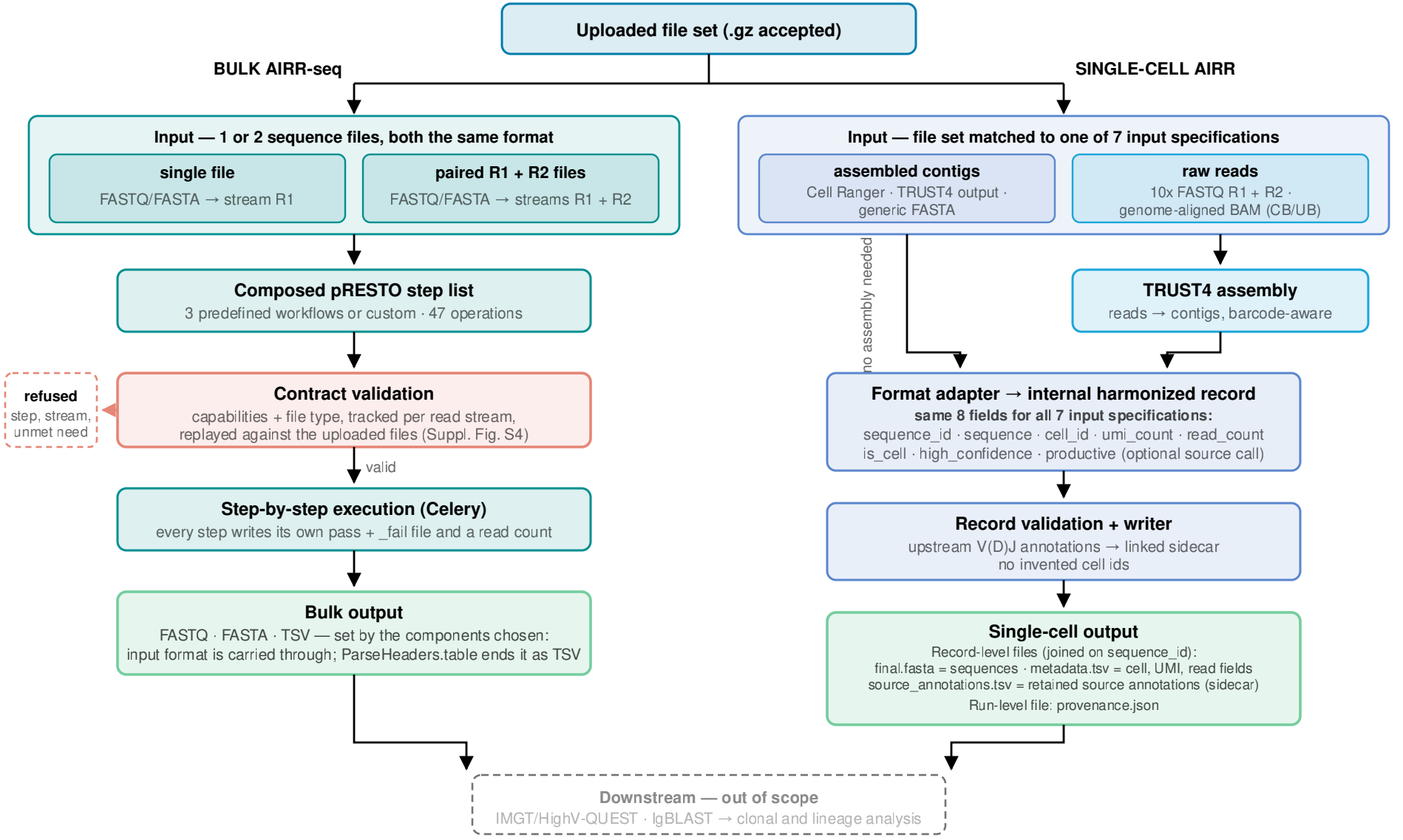

Uploaded file set (.gz accepted)
BULK AIRR-seq
SINGLE-CELL AIRR
Input — 1 or 2 sequence files, both the same format
single file
FASTQ/FASTA → stream R1
paired R1 + R2 files
FASTQ/FASTA → streams R1 + R2
Input — file set matched to one of 7 input specifications
assembled contigs
Cell Ranger · TRUST4 output · generic FASTA
raw reads
10x FASTQ R1 + R2 · genome-aligned BAM (CB/UB)
Composed pRESTO step list
3 predefined workflows or custom · 47 operations
no assembly needed
TRUST4 assembly
reads → contigs, barcode-aware
refused
step, stream, unmet need
Contract validation
capabilities + file type, tracked per read stream, replayed against the uploaded files (Suppl. Fig. S4)
valid
Format adapter → internal harmonized record
same 8 fields for all 7 input specifications:
sequence_id · sequence · cell_id · umi_count · read_count
is_cell · high_confidence · productive (optional source call)
Step-by-step execution (Celery)
every step writes its own pass + _fail file and a read count
Record validation + writer
upstream V(D)J annotations → linked sidecar
no invented cell ids
Bulk output
FASTQ · FASTA · TSV — set by the components chosen:
input format is carried through; ParseHeaders.table ends it as TSV
Single-cell output
Record-level files (joined on sequence_id):
final.fasta = sequences · metadata.tsv = cell, UMI, read fields
source_annotations.tsv = retained source annotations (sidecar)
Run-level file: provenance.json
Downstream — out of scope
IMGT/HighV-QUEST · IgBLAST → clonal and lineage analysis